AHANAF MOKAMMEL OMI

Full-Stack Developer
Dhaka-1205, Bangladesh | +880 1711-581367 | ahanafmo@gmail.com | Portfolio
github.com/BigSmoke4 | linkedin.com/in/ahanaf-mokammel-omi

Summary

Full-Stack Developer experienced in building scalable web applications and backend systems with ASP.NET Core, C#, SQL Server, PostgreSQL, Redis, and JavaScript. Strong in system design, database optimization, distributed systems, and AI-powered applications, with a focus on clean code, performance, and reliable software delivery.

Experience

Full-Stack Developer **2025 – Present**
Billing Master Software Ltd., Elephant Road, Dhaka-1205, Bangladesh
- Built 3+ ASP.NET MVC modules, improving internal processing efficiency by 30%.
- Implemented scalable CRUD and database features that reduced user-reported errors by 25%.
- Optimized SQL queries resulting in 20% faster data retrieval.

Executive Member **2022 – 2025**
BRAC University Computer Club, Merul Badda, Dhaka-1212, Bangladesh
- Organized university-wide coding workshops and hackathons with 100+ student participants.
- Collaborated with the web team to maintain club website and event registration systems.

Projects

NEXUS – Supply Chain Intelligence Command Center | *ASP.NET Core MVC, C#, PostgreSQL, Redis, EF Core, Docker* GitHub
- Engineered a supply-chain decision-support platform modeling dependencies as graphs: BFS/DFS, alternate paths, SPOF/bottleneck detection, impact propagation.
- Built deterministic disruption simulation (stockout-day, service-level, revenue-at-risk, what-if), AI orchestration, RBAC, audit logging, SignalR, Testcontainers.

VERITAS – Enterprise Identity & Access Governance Platform | *ASP.NET Core, C#, PostgreSQL, Redis, EF Core, Identity* GitHub
- Built a zero-trust IAM platform: RBAC, ABAC, deny-by-default ALLOW/DENY/APPROVAL evaluation, immutable policy versions, risk scoring, privileged access.
- Implemented risk-aware MFA (IP, login failure, after-hours, device-trust signals), JIT access, access reviews, API-key hashing/rotation, HMAC-SHA256 webhooks.

NEURA – Neural AI Agent Orchestration & Continuity Platform | *ASP.NET Core, C#, PostgreSQL, Redis, SignalR, OpenAI, Anthropic, Google AI* GitHub
- Developed a multi-agent platform maintaining task continuity across AI providers via ContextHandoffPackage, validation, automatic handoff, weighted routing.
- Added memory, knowledge graphs, AI adapters, retry/backoff, circuit breakers, rate limiting, encrypted credentials, cost tracking, SignalR/Cytoscape visualization.

FORGE – AI Software Engineering Intelligence Platform | *ASP.NET Core, C#, Roslyn, PostgreSQL, EF Core, Git* GitHub
- Engineered an intelligence pipeline combining Roslyn semantic analysis, repository indexing, C# symbol extraction, dependency graphs, and Git-history mining.
- Implemented deterministic risk scoring, security-pattern and technical-debt analysis, architecture policies, multi-tenant authZ, audit logging, rate limiting.

PULSE – Distributed Job Processing & Workflow Platform | *ASP.NET Core MVC, C#, RabbitMQ, Redis, SQL Server, EF Core* GitHub
- Engineered a distributed job-processing modular monolith: RabbitMQ priority queues, manual acks, dead-letter queues, Redis locks, idempotency, worker heartbeats.
- Implemented job lifecycle management, retries with exponential backoff, cancellation, scheduled jobs, worker orchestration, queue monitoring, failure simulation.

BDPRICEFINDER – Multi-Vendor E-Commerce Platform | *ASP.NET MVC, Razor Views, C#, HTML, CSS, JavaScript, SQL Server* Website
- Developed a full-stack Bangladesh multi-vendor price-comparison marketplace with role-specific dashboards, vendor workflows, approvals, hierarchical permissions.
- Implemented product lifecycle, dynamic variants, inventory/orders, 2-step email OTP auth, coupons, EMI, warranty, Elasticsearch search, Redis caching.

LOCAL AGENT – Local-First Autonomous Coding Platform | *ASP.NET Core, C#, PostgreSQL, EF Core, Ollama, Roslyn, SignalR* GitHub
- Built local-first autonomous agent: persistent sessions, DAG task planning, dependency validation, fan-out/fan-in execution, retry budgets, verification loops.
- Implemented SHA-256 incremental repository indexing, Roslyn symbol extraction, cross-file reference graphs, layered memory with semantic/lexical retrieval.

ATLAS – Enterprise Intelligent API & Event Platform | *ASP.NET Core, C#, PostgreSQL, Redis, Kafka, OpenTelemetry* GitHub
- Engineered a modular platform with API management, service registry, traffic routing, Kafka event processing, retries, DLQ/replay, Redis Lua rate limiting.
- Added circuit breakers, tenant authZ, OpenTelemetry SLO/error-budget analysis, health probing, incident management, deployment regression analysis, audit controls.

AURORA – Organizational Resilience & Decision Intelligence Architecture | *ASP.NET Core MVC, C#, PostgreSQL, EF Core, SignalR* GitHub
- Developed an enterprise digital-twin platform: BFS/DFS, Dijkstra, upstream/downstream propagation, SPOF/bottleneck scoring, disruption simulation.
- Added weighted risk scoring, AI investigation, human-in-the-loop approvals, audit trails, multi-tenancy; separated implemented from planned RAG/multi-agent features.

SENTINEL – Enterprise Operational Risk & Resilience Simulator | *ASP.NET Core MVC, C#, PostgreSQL, Redis* GitHub
- Built operational-risk simulator: dependency graphs, BFS/DFS, SPOF detection, critical-path analysis, weighted failure propagation, business-impact modeling.
- Added genuine Monte Carlo simulation (normal/log-normal), recovery-strategy scoring, resilience metrics, risk heatmaps, background jobs, graph visualization.

AEGIS – Intelligent Incident Response & Digital Operations Platform | *ASP.NET Core MVC, C#, PostgreSQL, Redis* GitHub
- Developed an SRE/AIOps platform with a synthetic infrastructure digital twin: Failure → Detection → Correlation → Root Cause → Impact → Remediation.
- Implemented rolling z-score anomaly detection, graph-based incident correlation, weighted root-cause ranking, business-impact analysis, human-approved runbooks.

NIRBHOR – Bilingual AI Personal Administrative Agent | *ASP.NET Core MVC, C#, PostgreSQL, EF Core, Identity, SignalR, Anthropic* GitHub
- Built a bilingual AI agent (English, Bangla, mixed-language) for tasks such as NID correction and passport renewal via conversation and tool execution.
- Implemented versioned prompt management, multilingual evaluation datasets, accuracy-gated evaluation, audit logging, notifications, cost tracking, OpenTelemetry.

E-COMMERCE – ASP.NET MVC/Razor Web Application | *ASP.NET MVC, C#, Razor Views, Entity Framework, SQL Server* GitHub
- Developed responsive database-driven e-commerce app with separated Controllers, Models, Services, Views, supporting product management and shopping cart.

Skills

Backend & Web: C#, ASP.NET Core/MVC, Razor, REST APIs, SignalR, Python, JavaScript, HTML5, CSS3.
Data & Infrastructure: SQL Server, PostgreSQL, Entity Framework Core, Redis, Elasticsearch, RabbitMQ, Kafka, Docker, Git/GitHub.
Architecture & Engineering: System Design, Modular Monolith, Dependency Injection, Event-Driven Design, Caching, Query Optimization, Authentication/RBAC, OpenTelemetry, Testing (k6).
AI Integration: OpenAI, Anthropic, and Google AI APIs, Multi-Agent Orchestration, Prompt Management.

Certificates

CS50's Introduction to Artificial Intelligence with Python (Harvard University), **Software Engineer Role** (HackerRank), **Neo4j & Generative AI Certification** (Neo4j), **Foundational C# Developer** (Microsoft & freeCodeCamp), **ASP.NET Core Fundamentals** (Microsoft), **Legacy Responsive Web Design V8** (freeCodeCamp), **Python Essentials 2**, **JavaScript Essentials 1**, **CSS Essentials**, **English for IT 2 – B2** (all Cisco Networking Academy), **Google Analytics** (Google), **HubSpot SEO II** (HubSpot Academy). "'

Education

B.Sc. in Computer Science – BRAC University, Kha 224 Pragati Sarani, Merul Badda, Dhaka-1212, Bangladesh. **2026**
Higher Secondary Certificate – Dhaka Residential Model College, Mirpur Road, Mohammadpur, Dhaka-1207, Bangladesh. **2020**
Secondary School Certificate – Ibn Taimiya School and College, Cumilla-3500 Bangladesh. **2018**